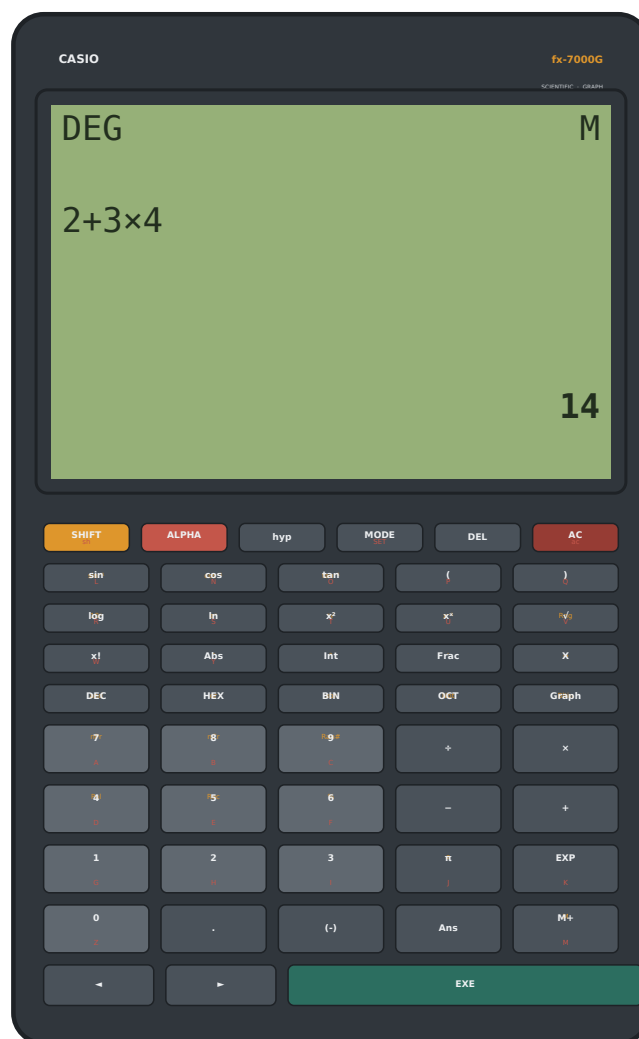


CASIO

fx-7000G Scientific & Graphing Calculator

Owner's Manual

Operation guide · Worked examples · Key reference



A faithful recreation of the 1985 CASIO fx-7000G — Kotlin + Jetpack Compose edition.

Dot-matrix 96 × 64 LCD · DEG/RAD/GRA · BASE-n · function graphing.

Preface

Congratulations on your fx-7000G. This calculator revives the spirit of the machine that, in 1985, became the world's first graphing scientific calculator — now rebuilt as a modern Android application with a genuine dot-matrix display, a full scientific engine and function graphing.

This manual is your complete guide. It explains every key, every operating mode and every function, and it is filled with worked examples set out in the classic “Operation / Display” style: read the key sequence on the left, and see exactly what appears on the LCD on the right. Follow along on your own calculator and you will master it in an afternoon.

The examples in this book are not decorative — each one is a real calculation the engine performs, with the exact result the display produces. Where a key has more than one meaning, the manual shows how the SHIFT, ALPHA and hyp prefixes unlock its second and third functions.

How to use this manual

- If you have never used the calculator before, read Chapter 1 (General Guide) first — it explains the display and the prefix keys that everything else depends on.
- For everyday arithmetic and editing, see Chapter 2.
- Use the Table of Contents (overleaf) to jump to a topic, and the alphabetical Index at the back to look up a specific function or key.
- Throughout, a key on the keypad is drawn as a chip, e.g. the EXE key, and multi-key operations are shown as a sequence.

Reading the symbols: In this manual the display glyphs are printed exactly as they appear on the calculator: $\times$ and $\div$ for multiply and divide, $-$ for subtract/negative, $\sqrt{}$ for square root, π for pi and $\rightarrow$ for the variable-store arrow.

Contents

1 General Guide	5
1-1 The keyboard	5
1-2 The prefix keys: SHIFT, ALPHA, hyp	5
1-3 The display	6
1-4 Turning entries into answers	6
2 Basic Calculations	7
2-1 Arithmetic and precedence	7
2-2 Negative numbers	7
2-3 Implicit multiplication	7
2-4 Answer memory (Ans)	7
2-5 Editing: replay, DEL and AC	8
3 Modes and Display Formats	9
3-1 Angle unit: DEG / RAD / GRA	9
3-2 Display format: Norm / Fix / Sci	9
4 Scientific Function Calculations	10
4-1 Trigonometric functions	10
4-2 Hyperbolic functions	10
4-3 Logarithms and exponentials	10
4-4 Powers, roots and reciprocals	10
4-5 Factorial, percent and combinatorics	11
4-6 Number functions: Abs, Int, Frac	11
4-7 Coordinate conversion: Pol and Rec	11
4-8 Degrees, minutes, seconds	11
5 Memory and Variables	13
5-1 Variables A–Z	13
5-2 Independent memory (M+ and M)	13
6 BASE-n Calculations	14
6-1 Choosing a base	14
6-2 Bitwise logic	14
7 Graphing Functions	15
7-1 Drawing a graph	15
7-2 The graph window (Range)	15
7-3 Built-in graphs (Bltin)	15
7-4 Tracing a curve	15
8 Error Messages	17
Appendix A Key Reference	18
Appendix B Specifications	19

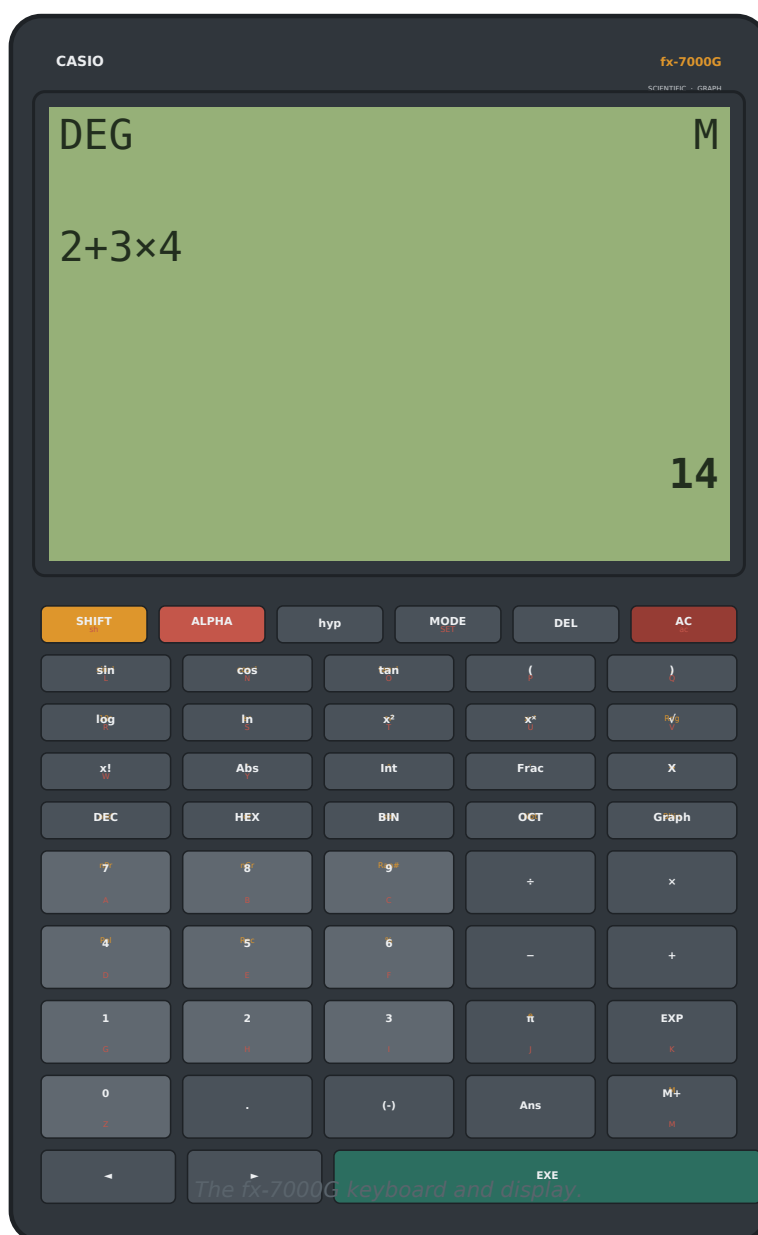
CHAPTER 1

General Guide

This chapter introduces the parts of the calculator you will use in every calculation: the display, the keyboard, and the three prefix keys that give each button more than one function.

1-1 The keyboard

The keyboard is arranged in ten rows. The upper five rows are the slim function keys (scientific functions, modes and base conversion); the lower rows carry the numbers, the four arithmetic operators and the large EXE key that evaluates an entry. Most keys print a small legend above and below the main label: the upper (orange) legend is reached with SHIFT, the lower (red) legend with ALPHA.



1-2 The prefix keys: SHIFT, ALPHA and hyp

Three keys do not enter anything by themselves. Instead they change the meaning of the next key you press. They are one-shot: after the next key they switch off automatically. While a prefix is active a small indicator (S, A or h) appears at the right of the status line.

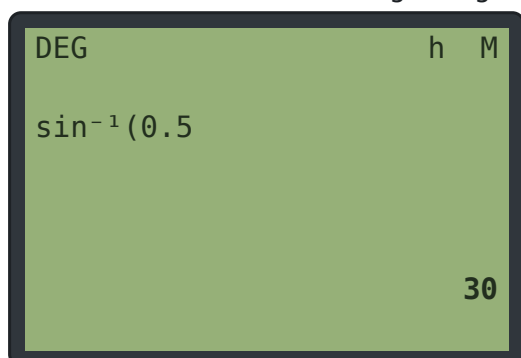
- SHIFT selects the function printed in orange above a key. Example: SHIFT then the sin key gives $\sin^{-1}$ (arc-sine).
- ALPHA selects the red letter printed below a key, so you can enter the variables A through Z.
- hyp turns the sin / cos / tan keys into the hyperbolic functions sinh, cosh and tanh. Press SHIFT before hyp to reach the inverse hyperbolics $\sinh^{-1}$, $\cosh^{-1}$ and $\tanh^{-1}$.

Note: SHIFT and ALPHA are mutually exclusive — turning one on turns the other off. Pressing a prefix a second time cancels it.

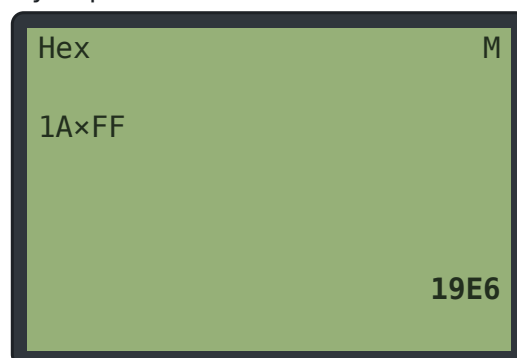
1-3 The display

The screen is a dot-matrix liquid-crystal display of 96×64 pixels. Text is drawn on a 5×7 pixel font inside 6×8 cells, giving 16 characters across and up to 8 rows. In calculation the layout is:

- Status line (top): the current mode at the left (DEG, RAD, GRA, or the base name Hex/Oct/Bin/Dec), and at the right the S / A / h prefix flags and the M memory flag.
- Entry line: the expression you are typing, over two rows if needed (up to 32 characters), with the cursor kept in view.
- Result line: the answer, right-aligned, shown after you press EXE.



Status flags, an entry line and a right-aligned result.



BASE-n mode: the active base replaces the angle unit.

1-4 Entering and evaluating

Type an expression using the number and function keys, then press EXE to evaluate it. The result appears on the result line and is remembered as Ans (see Chapter 2). To correct a mistake use DEL to erase the character before the cursor, or AC to clear the whole entry.

2 → + → 3 → × → 4 → EXE

The sequence above computes $2 + 3 \times 4$. Because $\times$ binds tighter than $+$, the answer is 14 (not 20).

CHAPTER 2

Basic Calculations

2-1 Arithmetic and precedence

The four operators $+$, $-$, $\times$ and $\div$ work as you expect. Multiplication and division are evaluated before addition and subtraction; use parentheses to override the order. The closing parenthesis may be omitted at the end of an expression — the calculator supplies it automatically when you press EXE.

Example 1 — order of operations

Operation	Display
$2+3\times 4$ EXE	14
$(2+3)\times 4$ EXE	20
$2\times(3+4)$ EXE	14

The last line shows the optional closing parenthesis: $2\times(3+4$ is read as $2\times(3+4)$.

2-2 Negative numbers

Use the $(-)$ key to enter a negative value, and the $-$ key for subtraction. Both print the same minus glyph on the display; the engine understands a leading minus as negation.

Example 2 — signed values

Operation	Display
$(-)5+8$ EXE	3
$8-(-)5$ EXE	13

2-3 Implicit multiplication

You may leave out the $\times$ sign in front of a constant, a variable, an opening parenthesis or a function. The calculator inserts the multiplication for you. This makes expressions read like ordinary mathematics.

Example 3 — omitting the multiply sign

Operation	Display
2π EXE	6.283185307
$2(3+2)$ EXE	10
$(2)(3)$ EXE	6

2-4 Answer memory (Ans)

The result of every calculation is stored in Ans. Press the Ans key to insert it into a new expression. As a shortcut, if you press an operator immediately after EXE, the calculator begins the new entry with Ans automatically — so you can chain calculations.

Example 4 — chaining with Ans

Operation	Display
150×1.05 EXE	157.5
+ (auto Ans)	Ans+
10 EXE	167.5
Ans÷2 EXE	83.75

2-5 Editing an entry

Move the cursor with the ◀ and ▶ keys. Pressing an arrow just after EXE replays the previous expression so you can edit and re-run it. DEL deletes the item to the left of the cursor — a whole function name such as sin(or Ans is removed in a single press. AC clears everything and also erases any graph.

- ◀ / ▶ move the cursor (or, right after EXE, recall the last entry for editing).
- DEL erase one character or one whole token to the left.
- AC clear the entry, the result and any graph curves.

CHAPTER 3

Modes and Display Formats

3-1 Angle unit — DEG / RAD / GRA

The MODE key cycles the angle unit used by the trigonometric functions: DEG → RAD → GRA → DEG. The active unit is always shown at the left of the status line. In degrees a right angle is 90, in radians $\pi/2$, and in grades 100.

Example 5 — the same angle in three units

Operation	Display
[DEG] sin30 EXE	0.5
[RAD] sin($\pi \div 2$) EXE	1
[GRA] sin100 EXE	1

3-2 Display format — Norm / Fix / Sci

Press SHIFT then MODE (the SET function) to open the setup menu. From here you choose the angle unit or the number-display format. The menu offers:

- 1 Deg 2 Rad 3 Gra — set the angle unit directly.
- 4 Fix — then a digit 0–9 fixes that many decimal places.
- 5 Sci — then a digit 1–9 sets the number of significant figures (0 means 10).
- 6 Norm — return to the standard 10-significant-digit display.

In Norm the calculator shows up to 10 significant digits, switching to exponential notation (mantissa E exponent) for magnitudes of 10^{10} or greater, or smaller than 10^{-10} . Fix rounds to a fixed number of decimals; Sci always uses exponential form.

Example 6 — the same value in three formats

Operation	Display
[Norm] 2÷3 EXE	0.6666666667
[Fix 4] 2÷3 EXE	0.6667
[Sci 3] 2÷3 EXE	6.67E-1

Set the format from SHIFT → MODE (SET) before evaluating.

Opening SET: SET is reached with SHIFT → MODE. After choosing 4 Fix or 5 Sci the display prompts for the digit count — just press a number key.

CHAPTER 4

Scientific Function Calculations

4-1 Trigonometric and inverse functions

The sin, cos and tan keys open a function with a left parenthesis, e.g. sin(. They use the current angle unit. Their SHIFT functions $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$ are the inverse (arc) functions.

Example 7 — trig and arc-trig (DEG)

Operation	Display
cos60 EXE	0.5
tan45 EXE	1
$\sin^{-1}(0.5)$ EXE	30

4-2 Hyperbolic functions (hyp)

Press hyp before sin, cos or tan to get sinh, cosh and tanh. Press SHIFT then hyp then the key for the inverse hyperbolic functions.

hyp → sin → 1 → EXE

Example 8 — hyperbolic

Operation	Display
hyp sin1 EXE	1.175201194
hyp cos0 EXE	1
SHIFT hyp tan(0.5) EXE	0.5493061443

4-3 Logarithms and exponentials

log is the common (base-10) logarithm and ln is the natural logarithm. Their SHIFT functions are the antilogarithms: 10^x and e^x , which begin the power expressions $10^{\wedge}()$ and $e^{\wedge}()$.

Example 9 — logs and powers of the base

Operation	Display
log100 EXE	2
ln(e EXE	1
$10^{\times}3$ EXE	1000
$e^{\times}1$ EXE	2.718281828

4-4 Powers, roots and reciprocals

The calculator offers x^2 (square), x^x (general power, the $\wedge$ key), $\sqrt{}$ (square root), $\sqrt[n]{}$ (the index-th root, SHIFT of x^2), and x^{-1} (reciprocal, SHIFT of the power key). Powers are evaluated right-to-left, so 2^3^2 is $2^{(3^2)} = 512$.

Example 10 — powers and roots

Operation	Display
5^2 EXE	25
2^{10} EXE	1024
2^{3^2} EXE	512
$\sqrt[3]{2}$ EXE	1.414213562
$3 \times \sqrt{8}$ EXE	2
4^{-1} EXE	0.25

$3 \times \sqrt{8}$ is the cube root of 8; the index is entered before the $\times \sqrt{}$ symbol.

4-5 Factorial, percent, nPr and nCr

$x!$ gives the factorial of a whole number (0 to 69). % divides by 100. nPr and nCr (SHIFT of 7 and 8) are the permutation and combination counts; they bind more tightly than + and – but looser than $\times$ and $\div$.

Example 11 — counting

Operation	Display
$5!$ EXE	120
50% EXE	0.5
$5nPr2$ EXE	20
$5nCr2$ EXE	10

4-6 Abs, Int and Frac

Abs(returns the absolute value. Int(truncates toward zero to the integer part, and Frac(returns the fractional part (the value minus its integer part).

Example 12 — parts of a number

Operation	Display
Abs((-)7 EXE	7
Int(3.7 EXE	3
Frac(3.7 EXE	0.7
Int((-)3.7 EXE	-3

4-7 Coordinate conversion — Pol and Rec

Pol(x, y) converts rectangular coordinates to polar: it returns the radius r and stores the angle Π in variable J (with r also placed in I). Rec(r, θ) does the reverse, returning x and storing y in J. The comma is the SHIFT of the) key. The angle uses the current unit.

Example 13 — rectangular → polar (DEG)

Operation	Display
Pol(3,4 EXE	5
J EXE	53.13010235
Rec(5,53.13 EXE	3

After Pol, recall the angle by pressing ALPHA then J. Results I and J are ordinary variables.

4-8 Sexagesimal (degrees ° minutes ' seconds ")

Enter an angle in degrees-minutes-seconds using the ° (SHIFT Int), ' (SHIFT Frac) and " (SHIFT Abs) marks. The calculator converts the entry to decimal degrees.

Example 14 — DMS to decimal degrees

Operation	Display
30°30' EXE	30.5
1°30'36" EXE	1.51

CHAPTER 5

Memory and Variables

5-1 Variables A through Z

The calculator has 26 lettered memories, A to Z. Enter a letter with the ALPHA prefix. Store a value by following it with the $\rightarrow$ arrow (SHIFT of the X key) and a letter: value $\rightarrow$ A. Recall a variable simply by using its letter in an expression; an unset variable reads as 0.

Example 15 — storing and recalling

Operation	Display
5 $\rightarrow$ A EXE	5
A $\times$ 2 EXE	10
A ² +1 EXE	26

Type A with ALPHA $\rightarrow$ (the 7 key's red legend), and $\rightarrow$ with SHIFT $\rightarrow$ X.

5-2 Independent memory — M+ and M

The M+ key adds the current value to the independent memory M. Recall M with SHIFT M+ (the M legend). Whenever M holds a non-zero value the M flag is shown on the status line. Clear it by storing 0 into M (0 $\rightarrow$ M).

Example 16 — accumulating a running total

Operation	Display
250 M+	250
120 M+	120
M EXE	370

Each M+ adds to M; recall the total with the M legend (SHIFT M+).

CHAPTER 6

BASE-n Calculations

6-1 Decimal, hexadecimal, binary and octal

The DEC, HEX, BIN and OCT keys switch the calculator into integer base-n mode and convert the current value to that base. The active base (Dec, Hex, Bin or Oct) is shown on the status line in place of the angle unit. In a non-decimal base the keypad accepts only the digits valid for that base — hexadecimal digits A-F are the ALPHA letters. Values are whole numbers only.

Example 17 — converting between bases

Operation	Display
[DEC] 255 HEX	FF
[same value] BIN	11111111
[same value] OCT	377
[same value] DEC	255

Press a base key to convert the shown value; press DEC to return to ordinary decimal.

6-2 Bitwise logic — and, or, xor, Not

In base-n mode the SHIFT legends of the base keys give the logical operators and, or, xor and Not, which act bit by bit on the integer operands. Their precedence, from loosest to tightest, is: or, then xor, then and, then + and −, then × and ÷, then the unary Not. These words also work in decimal, where they force integer arithmetic.

Example 18 — logic in hexadecimal / decimal

Operation	Display
[HEX] 1A and F EXE	A
[HEX] 1C or 3 EXE	1F
[DEC] 12 and 8 EXE	8
[DEC] Not 0 EXE	-1

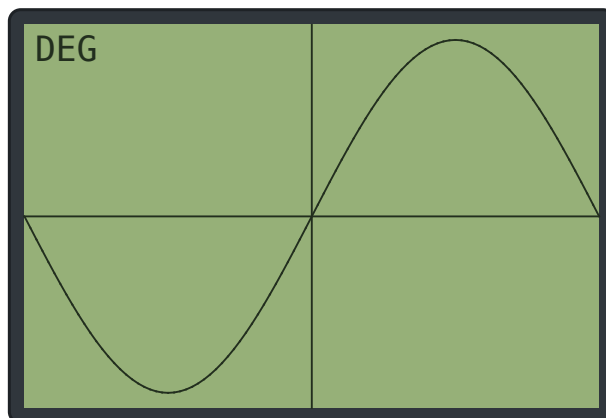
CHAPTER 7

Graphing Functions

7-1 Drawing $Y = f(X)$

Enter an expression that uses the variable X (the X key), then press Graph. The calculator plots $Y = f(X)$ across the display, drawing the axes with tick marks. Graph again with a new expression to overlay it on the previous curves — up to six curves are kept. Pressing any ordinary key returns you to the calculation screen.

sin → **X** → **Graph**



A plot of $Y = \sin X$ with axes and tick marks.

7-2 Setting the window — Range

Press SHIFT then $\sqrt{}$ (the Rng function) to open the Range editor. It lists six values that define the viewing window and the spacing of the tick marks. Move between fields with EXE (or the arrows), type a new value, and press Graph to redraw. The defaults are $XMIN=-4.7$, $XMAX=4.7$, $XSCL=1$, $YMIN=-3.1$, $YMAX=3.1$, $YSCL=1$.

- $XMIN / XMAX$ — the left and right edges of the window.
- $XSCL$ — the distance between tick marks along the x-axis.
- $YMIN / YMAX$ — the bottom and top edges.
- $YSCL$ — the tick spacing along the y-axis.

Note: If $XMAX$ is not greater than $XMIN$ (or $YMAX$ not greater than $YMIN$) the graph cannot be drawn and Ma ERROR is shown.

7-3 Built-in graphs — Bltin

Press SHIFT then Graph (the Bltin function) to choose a ready-made graph. Pick a number from the menu and the calculator plots it immediately:

- 1 $\sin X$ 2 $\cos X$ 3 $\tan X$ 4 X^2
- 5 X^3 6 $\sqrt{X}$ 7 $1/X$ 8 $\ln X$

7-4 Tracing

While a graph is shown, press ◀ or ▶ to switch on the trace cursor. A pointer moves along the most recent curve, and the X and Y coordinates of the cursor are read out at the foot of the screen. Keep pressing the arrows to move the pointer one dot-column at a time.

CHAPTER 8

Error Messages

When a calculation cannot be completed the display shows Ma ERROR. Press AC, DEL or any editing key to clear the message and correct the entry. The common causes are listed below.

Cause	Example
Division by zero	$5 \div 0$, or x^{-1} of 0
Argument outside a function's domain (e.g. $\log$ or $\sqrt{}$ of a non-positive number, $\sin^{-1}$ outside -1 – 1)	$\log 0$, $\sqrt{(-)4}$, $\sin^{-1}(2)$
Result too large to display (overflow, $ x \geq 10^{100}$; base-n beyond about 9.2×10^{18})	10×200
Factorial of a non-integer, a negative number, or a value above 69	$3.5!$, $70!$
nPr / nCr with invalid arguments (negative, non-integer, or $r > n$)	$3nPr5$
Incomplete or malformed expression	$2 + \times$

Appendix A — Key Reference

Every key with its primary function and its SHIFT (orange) and ALPHA (red) legends. Prefix keys (SHIFT, ALPHA, hyp) are one-shot.

Key	Primary function	SHIFT / ALPHA
SHIFT	select orange legend	—
ALPHA	select red letter	—
hyp	sinh/cosh/tanh prefix	—
MODE	cycle DEG/RAD/GRA	SET (Norm/Fix/Sci setup)
DEL / AC	delete token / clear all	—
sin cos tan	trig functions	$\sin^{-1}$ $\cos^{-1}$ $\tan^{-1}$ · ALPHA L N O
()	parentheses	)→ , (comma) · ALPHA P Q
log ln	base-10 / natural log	10^x e^x · ALPHA R S
x^2 x^x	square / power	$\sqrt[x]{}$ x^{-1} · ALPHA T U
$\sqrt{}$	square root	Rng (Range editor) · ALPHA V
x!	factorial	ALPHA W
Abs Int Frac	abs / integer / fraction	" ° ' · ALPHA Y
X	graph variable X	→ (store)
DEC HEX BIN OCT	select number base	and or xor Not
Graph	plot $Y = f(X)$	Bltn (built-in graphs)
7 8 9	digits	nPr nCr Ran# · ALPHA A B C
4 5 6	digits	Pol Rec % · ALPHA D E F
1 2 3	digits	ALPHA G H I
π EXP	pi / exponent (E)	e · ALPHA J K
0 . (-)	digit / point / sign	ALPHA Z
Ans	last answer	—
M+	add to memory M	M (recall) · ALPHA M
◀▶ EXE	cursor / evaluate	—

Appendix B — Specifications

Item	Specification
Display	Dot-matrix LCD, 96 × 64 pixels; 5×7 font, 16 characters × 8 rows
Entry line	Up to 32 characters over two rows, with a visible cursor
Number display	Norm (10 significant digits), Fix (0–9 decimals), Sci (1–10 sig. figures)
Overflow	Ma ERROR at $ x \geq 10^{100}$ (about 9.2×10^{18} in base-n)
Angle units	Degree, Radian, Gradient
Number bases	Decimal, Hexadecimal, Binary, Octal with and / or / xor / Not
Memories	Independent memory M; 26 variables A–Z; Ans; graph variable X
Functions	Trig & inverse, hyperbolic & inverse, log/lr, $10^x/e^x$, powers & roots, $x!$, %, nPr, nCr, Abs, Int, Frac, Pol, Rec, DMS
Graphing	$Y = f(X)$ plots, up to 6 overlaid curves, 8 built-in graphs, Range editor, trace
Constants	π , e, Ran# (random), Ans

Index

(		G	
(-) key	7	graph	15
1		Graph key	15
10^x	10	graph trace	16
A		graph window	15
A-Z	13	H	
Abs	11	HEX	14
AC key	8	hexadecimal	14
ALPHA key	6	hyp key	6
and	14	hyperbolic functions	10
angle mode	9	I	
Ans	7	implicit multiplication	7
answer memory	7	independent memory	13
arithmetic	7	Int	11
arrow key	13	integer part	11
B		inverse trigonometric functions	10
BASE-n	14	K	
BIN	14	key reference	18
binary	14	keyboard	5
bitwise	14	keys	5
Bltn	15	L	
built-in graphs	15	LCD	6
C		In	10
combination	11	log	10
coordinate conversion	11	logarithm	10
cos	10	logical operators	14
cosh	10	M	
cube root	10	M flag	13
cursor	8	M+	13
D		Ma ERROR	17
DEC	14	memory	13
DEG	9	MODE key	9
degrees minutes seconds	12	N	
DEL key	8	nCr	11
display	6	negative numbers	7
display format	9	Norm	9
DMS	12	Not	14
E		nPr	11
e^x	10	O	
error	17	OCT	14
exponential	10	octal	14
F		operator precedence	7
factorial	11	or	14
Fix	9	P	
Frac	11	percent	11
fractional part	11	permutation	11
G		Pol	11
GRA	9	polar	11

P

power	10
precedence	7
prefix keys	6
preset graphs	15

R

RAD	9
Range	15
Rec	11
reciprocal	10
rectangular	11
replay	8

S

Sci	9
SET menu	9
sexagesimal	12
SHIFT key	6
sin	10
sinh	10
specifications	19
square	10
square root	10
store	13

T

tan	10
tanh	10
trace	16
trigonometric functions	10

V

variables	13
-----------------	----

X

X variable	15
x^y	10
XMAX	15
XMIN	15
xor	14
XSCL	15